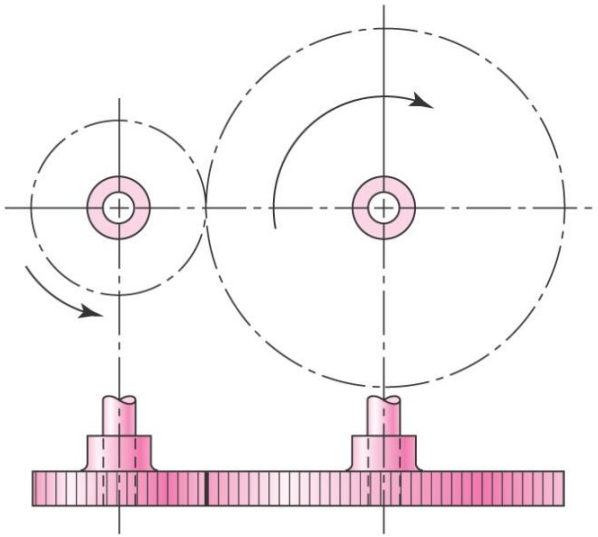
DESIGN OF SPUR GEARS

- REVIEW – Geometry, Nomenclature, etc.
- GEAR FORCE ANALYSIS
- GEAR TOOTH FORCE ANALYSIS
- GEAR TOOTH BENDING STRESS
  - Basic Analysis
  - Refined Analysis
- GEAR TOOTH SURFACE DURABILITY/ANALYSIS
  - Basic Concept
  - Refined Analysis
- SPUR GEAR DESIGN PROCESS

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Types of Gears



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Nomenclature of Spur-Gear Teeth

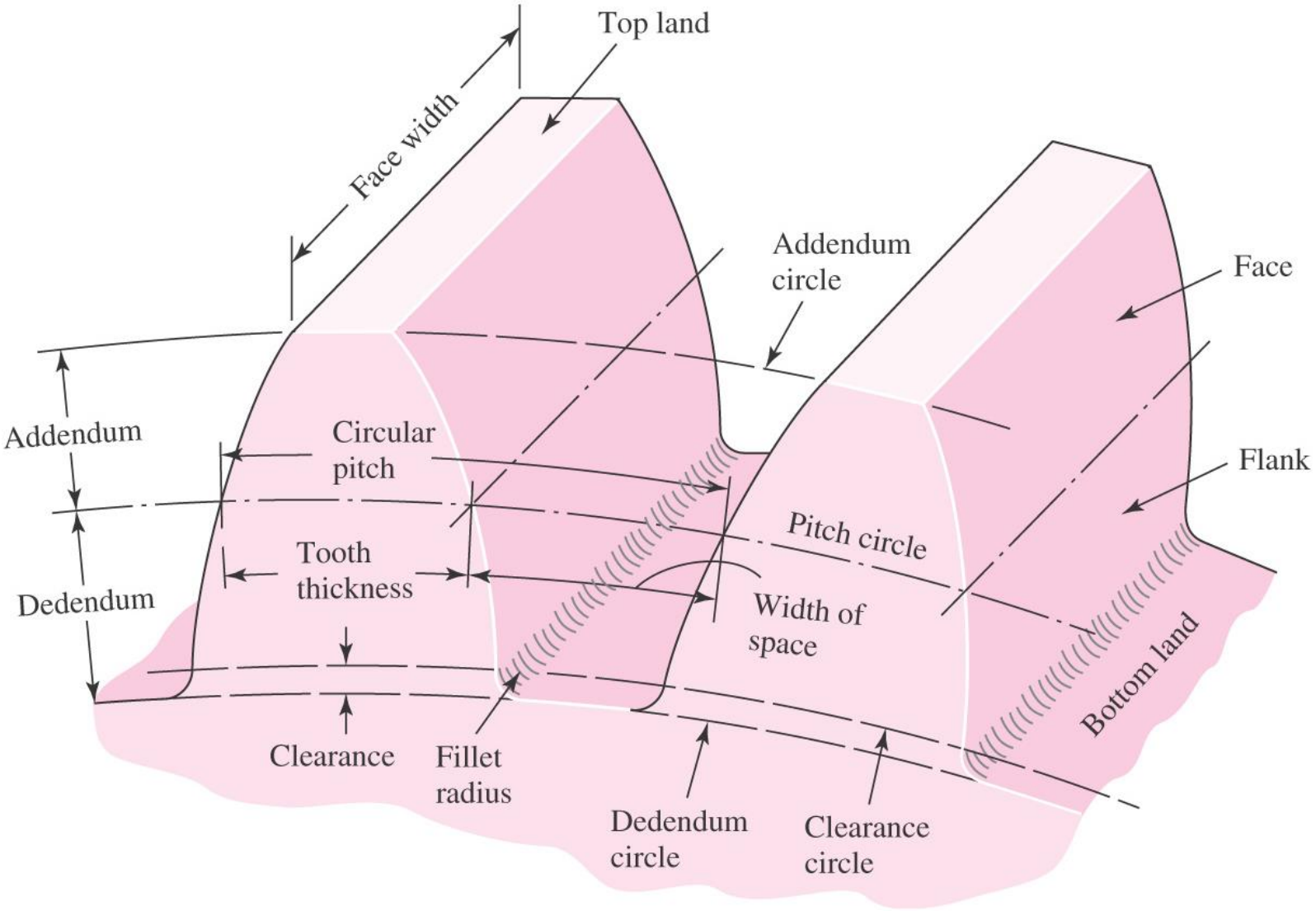


Fig. 13–5

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Gear Parameters

$$P = \frac{N}{d} \tag{13–1}$$

$$m = \frac{d}{N} \tag{13–2}$$

$$p = \frac{\pi d}{N} = \pi m \tag{13–3}$$

$$pP = \pi \tag{13–4}$$

where P = diametral pitch, teeth per inch
N = number of teeth
d = pitch diameter, in or mm
m = module, mm
p = circular pitch, in or mm

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Standardized Tooth Systems

- **Common Practical Face Width (F):**

$$3p < F < 5p$$

$$p = \frac{\pi}{P}$$

$$\frac{3\pi}{P} < F < \frac{5\pi}{P}$$

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Pressure Angle and Force Directions

- Common pressure angle ϕ: 20º and 25º
- Old pressure angle: 14 ½º **(DO NOT USE)**

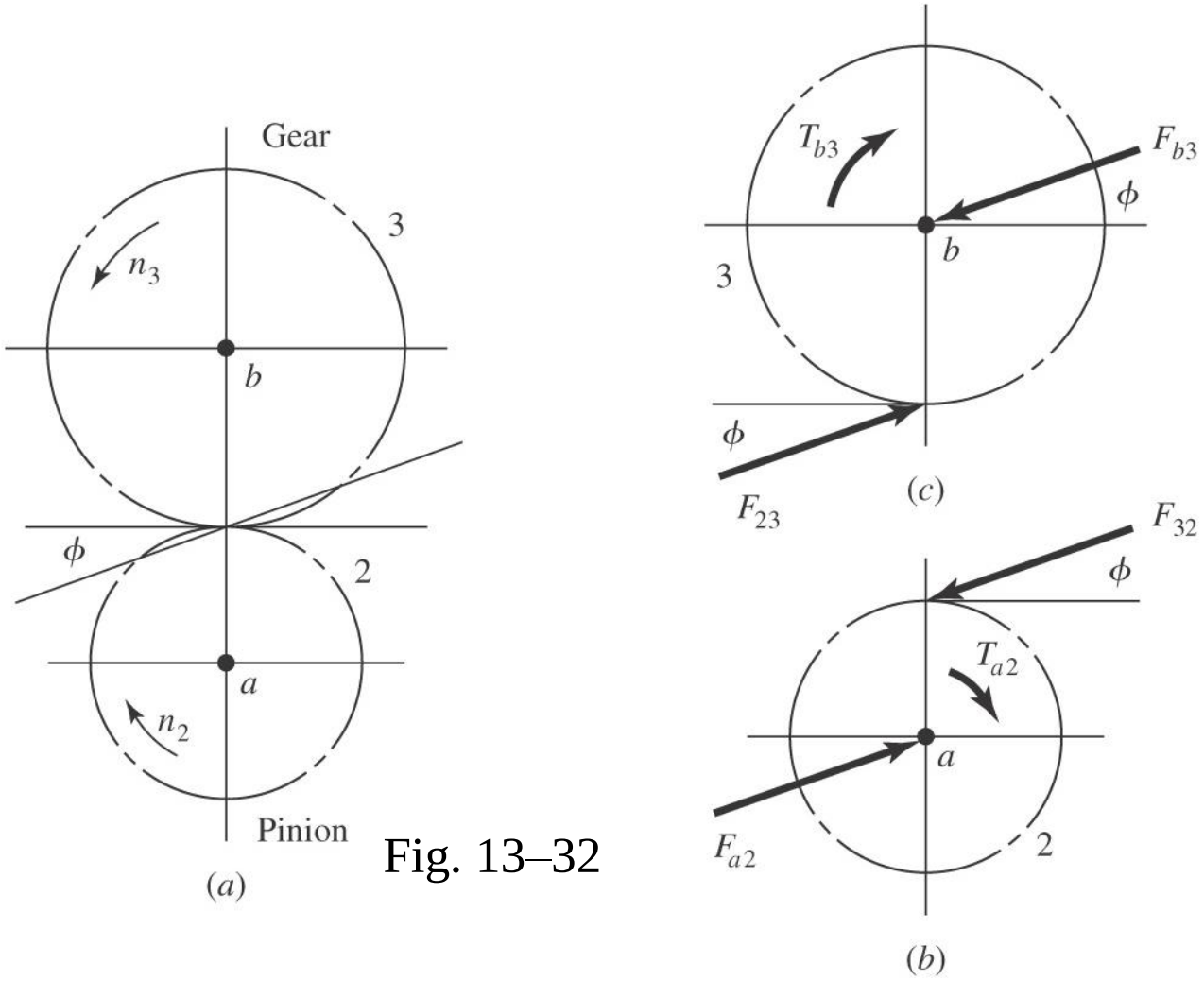


Fig. 13–32

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Force Analysis – Spur Gearing

- ***Transmitted load W_t*** is the tangential load

$$W_t = F_{32}^t$$

- It is the useful component of force, **transmitting the torque**

$$T = \frac{d}{2} W_t$$

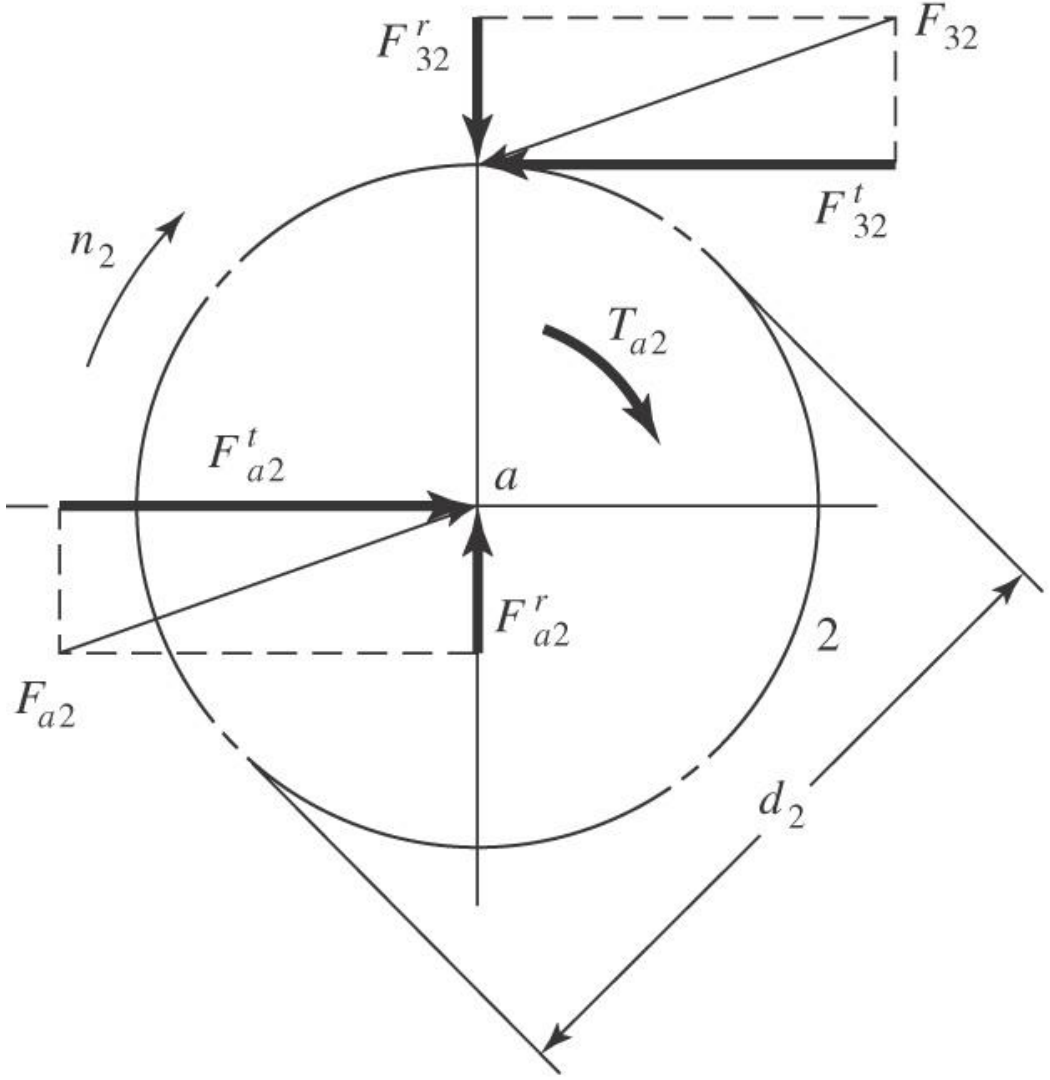


Fig. 13–33

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Power, Speed and Torque Relationship

- ***Transmitted power H***

$$H = T\omega = (W_t d/2)\,\omega \qquad (13\text{–}33)$$

- ***Pitch-line velocity*** is the linear velocity of a point on the gear at the radius of the pitch circle. It is a common term in tabulating gear data.

$$V = \pi dn/12 \qquad (13\text{–}34)$$

where V = pitch-line velocity, ft/min

d = gear diameter, in

n = gear speed, rev/min

SI Units: **V = πdn/1000**

V ---> m/s

d ---> mm

n ----> rev/s

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Transmitted Load

- Useful power relation in customary units,

$$W_t = 33\,000\frac{H}{V} \qquad (13\text{–}35)$$

where W_t = transmitted load, lbf

H = power, hp

V = pitch-line velocity, ft/min

- **In SI Units,**

$$W_t = \frac{60\,000H}{\pi dn} \qquad (13\text{–}36)$$

where W_t = transmitted load, kN

H = power, kW

d = gear diameter, mm

n = speed, rev/min

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Example 1 – Gear Force Analysis

Pinion 2 in Fig. 13–34*a* runs at 1750 rev/min and transmits 2.5 kW to idler gear 3. The teeth are cut on the 20° full-depth system and have a module of $m = 2.5$ mm. Draw a free-body diagram of gear 3 and show all the forces that act upon it.

- Find the total reaction force at center of gear 3
- Find the torque on each gear

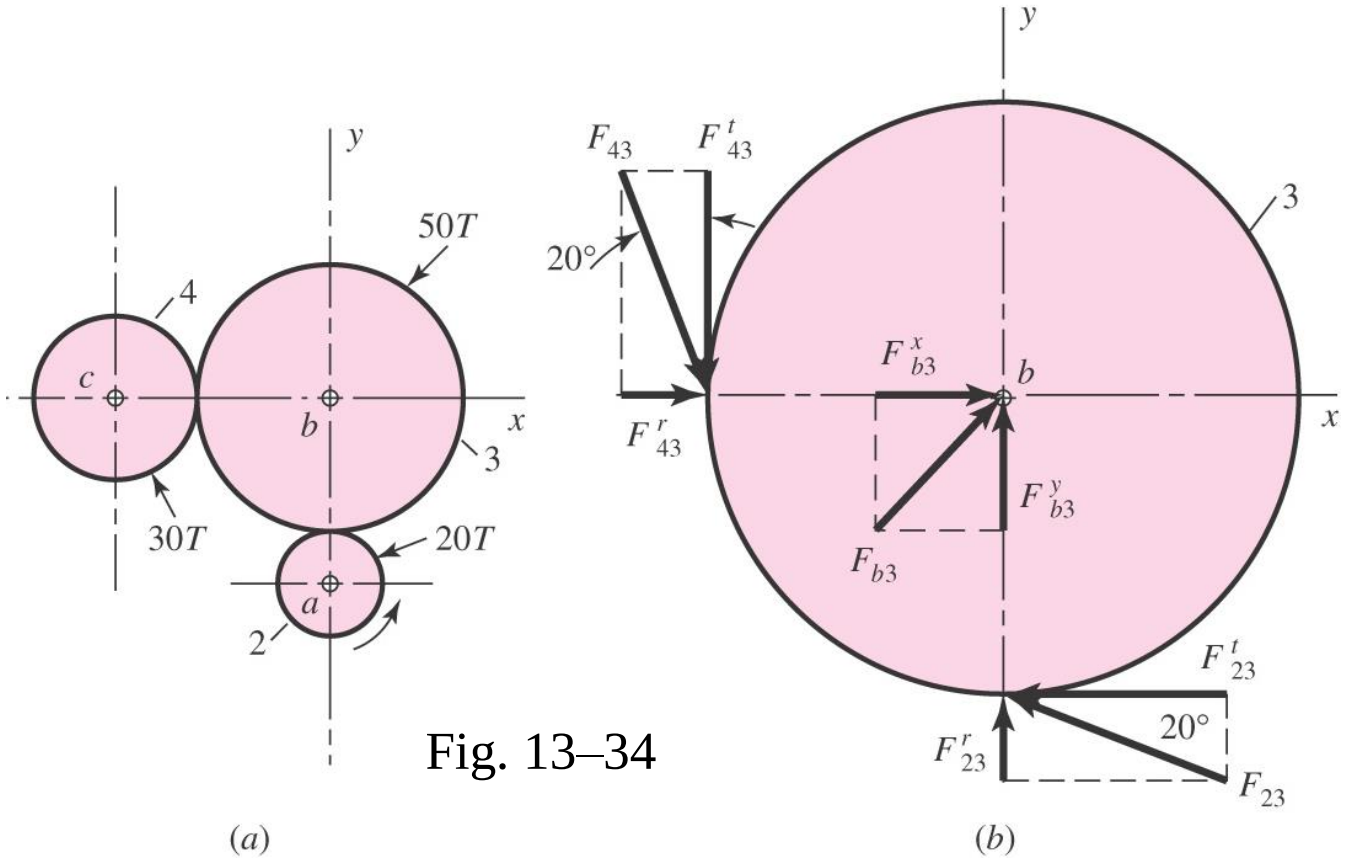
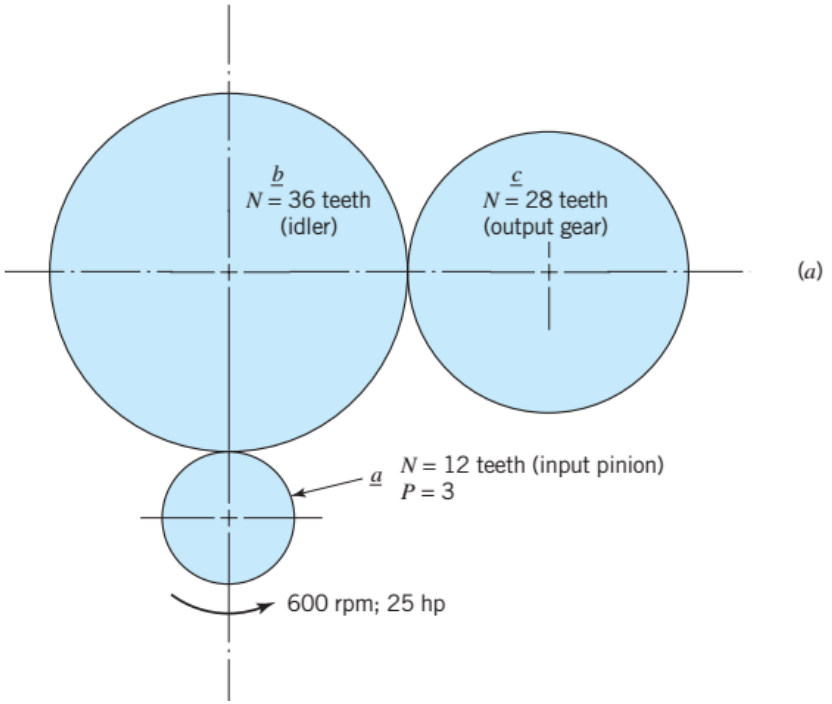


Fig. 13–34

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Practice Problem- Gear Force Analysis

Three gears of module m = 4 mm (diametral pitch P =3) and pressure angle of 20° are arranged as shown. Gear "a" is the driving or input pinion. It rotates Counterclockwise at 600 rpm and transmits 18 kW (25 hp) to idler gear "b". Output gear "c" is attached to a shaft that drives a machine. Nothing is attached to the idler gear and friction losses can be neglected.



a. Draw Free Body Diagram of Gear "b"

b. Determine the resultant load applied to the idler gear using SI Units.

c. Determine the resultant load applied to the idler gear using Customary Units.

d. Find the toque on each gear

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